



Disaster Recovery and Business Continuity Plan — TASCO Growth Platform

TASCO Insurance × VETC Growth Platform — Operations

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Document	Version	Date	Owner	Status
DR and BCP	1.0	07 October 2026	TASCO IT Head (accountable); SRE + DBA (responsible)	Issued for submission

Item	Value
Approvers	TASCO IT Head, TASCO Business Owner, VETC IT Lead, TASCO CISO
Related	Runbook and Support Guide · Monitoring and Alerting · Production Readiness Checklist
Hosting assumption	Hosting provider and regions are confirmed at G1. This plan assumes two data centres / regions in Vietnam (primary Hà Nội, secondary TP. Hồ Chí Minh) to meet data-localisation expectations for Vietnamese personal data (Decree 53/2022/ND-CP, PDP Law 2025; Legal confirms)

1. Scope

In scope	Out of scope (own DR plans)
Application tier (stateless Node.js pods), PostgreSQL database, secrets (JWT_SECRET, DATA_KEYS, BLIND_INDEX_KEY, DB credentials), configuration (rule sets live in the database , versioned), CronJobs, ingress/DNS, observability stack	VETC wallet, VETC SSO, VETC app, TASCO core policy admin, Zalo, SMS provider, voice-AI vendor. We plan for their unavailability (§8). The hosted UAT on Railway is a non-production environment and is not covered by this plan

Key architectural facts that make recovery simpler:

- **All state is in PostgreSQL.** This covers profiles, leads, touchpoints, messages, orders, policies, rule sets (all versions), the outbox (`domain_events`) and the hash-chained `audit_log` . The app pods are stateless, apart from per-pod rate-limiter buckets, the token revocation list and the 15 s rule cache, all of which are disposable.
- **Outbox pattern.** Events are written in the same database as the state change, so after a restore the relay re-processes `pending` events. Subscribers are idempotent, with the caveats in RB-06.
- **Encryption keys are part of the backup set.** A database backup is useless without the matching `DATA_KEYS` (every key ID ever active) and `BLIND_INDEX_KEY` . The keys live in the KMS/vault, replicated to the DR region.

2. Business impact analysis (BIA)

Business service	Components	Impact of outage	Tolerable downtime (MTPD)	Tier
Customer renewal / purchase (quote → pay → issue → e-certificate)	<code>/api/customer/*</code> , <code>/api/partner/v1/orders</code> , wallet, TASCO core	Lost premium (renewals on expiry day go to competitors); customer trust; driving uninsured is an offence for the customer	4 h	Tier 1
Certificate verification (QR)	GET <code>/api/public/certificates/:certNo</code> , <code>/verify/*</code>	Customers cannot prove cover at roadside or inspection checks	4 h	Tier 1
Partner API	<code>/api/partner/v1/*</code>	Partner sales stop; SLA penalties (contractual)	4 h	Tier 1

Business service	Components	Impact of outage	Tolerable downtime (MTPD)	Tier
Telesales console and handoffs	/api/handoffs, /api/leads, /api/customers/:id, /api/quotes	Hot leads go cold; agents idle	8 h	Tier 2
Journey engine and messaging	journeys CronJob, notify gateways, voice bot	Reminders delayed by a day; recoverable (catch-up of 2 days in planning, catchUpDays)	24 h	Tier 2
Claims FNOL	/api/customer/claims, /api/claims	Customers use the TASCO hotline instead	24 h	Tier 2
Rules studio, data steward, DQ, ingestion	/api/rules, /api/dq, /api/data/ingest	Changes deferred; ingestion re-runnable from source extracts	72 h	Tier 3
Dashboards, insights	/api/dashboard/*	Management reporting delayed	72 h	Tier 3
Audit trail (integrity)	audit_log	Regulatory evidence; must never be lost	n/a: zero tolerated loss of committed entries within the RPO	Tier 1 (data)

3. RPO / RTO by tier

Tier	RTO (service restored)	RPO (max data loss)	Mechanism
Tier 1	1 h for an AZ failure (automatic); 4 h for a region loss (declared DR)	≤ 5 min for a region loss; 0 for an AZ failure	Multi-AZ Postgres with a synchronous standby; async cross-region replica; continuous WAL archiving (PITR)
Tier 2	8 h	≤ 15 min	Same database. Journeys catch up the next run (catch-up window 2 days)
Tier 3	72 h	≤ 24 h	Same database; ingestion re-runnable from source extracts

Secrets/KMS: RTO 1 h, RPO 0 (replicated vault). Observability: RTO 24 h (rebuild from IaC; metrics history loss accepted).

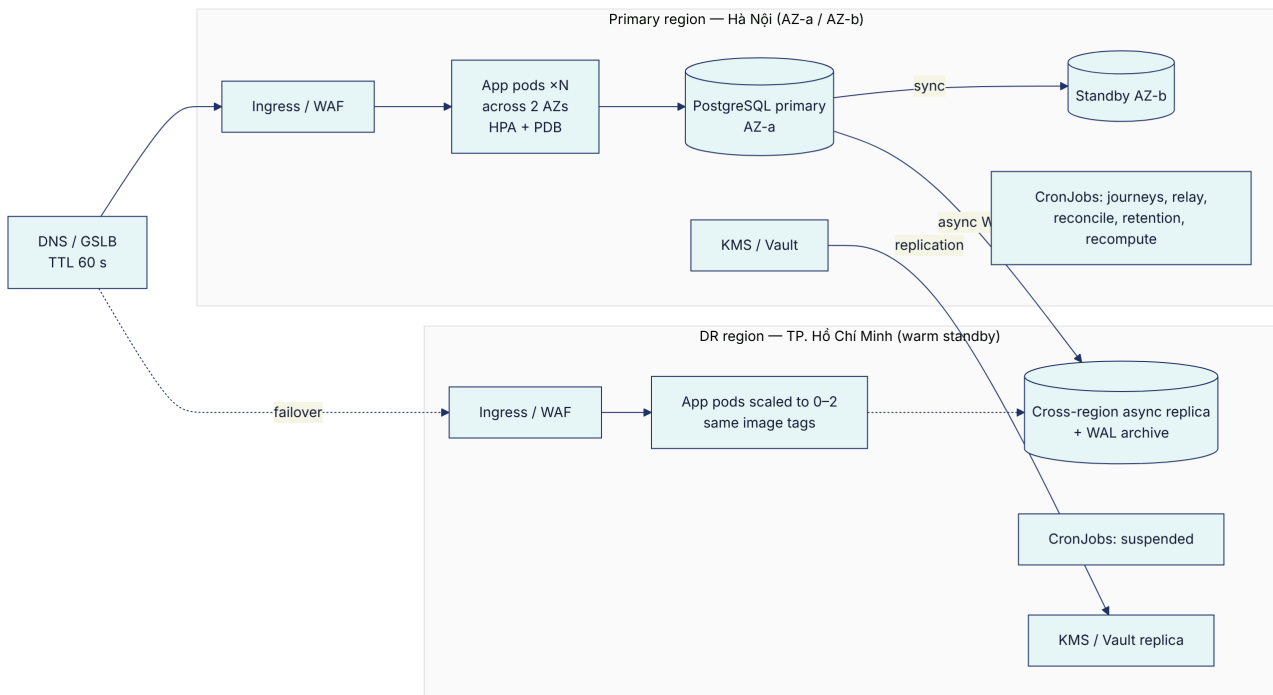
4. Backup strategy

Asset	Method	Frequency	Retention	Location	Verified by
PostgreSQL	Managed PITR : continuous WAL archiving + daily base backup	Continuous / daily 01:00 ICT	35 days PITR; monthly snapshot kept 12 months; yearly kept 10 years (financial and audit retention per retention.json: orders, certificates and audit_log 3,650 days)	Primary region object storage + cross-region copy (TP.HCM)	Weekly automated restore test (DR-T1 light)
audit_log chain head	Export (seq, hash) of the latest entry	Nightly	10 years	WORM bucket (object lock)	Compare with GET /api/audit/verify head

Asset	Method	Frequency	Retention	Location	Verified by
Secrets (JWT_SECRET, DATA_KEYS all key IDs, BLIND_INDEX_KEY, DB, partner-integration credentials)	Vault/KMS with replication + sealed offline escrow (two-person)	On change	All versions forever for DATA_KEYS (old ciphertext)	KMS in both regions + escrow	Quarterly key-recovery drill
Configuration / IaC	Git (deploy/k8s/*, Dockerfile, CI)	On commit	Forever	Git hosting + mirror	Rebuild drill (DR-T4)
Container images	Registry with immutable tags (semver + git SHA)	On release	Last 20 releases	Registry replicated to the DR region	Pull test in DR
Rule sets	Inside PostgreSQL (all versions) + defaults in config/rules/*.json	—	—	Covered by DB backup	GET /api/ops/status checksums vs change log
Source extracts (VETC/TASCO)	Landing bucket	Per delivery	365 days (source_records retention)	Primary + DR	Re-ingest test

Backups are encrypted at rest with provider KMS keys. **PII inside the backup is additionally encrypted at field level (AES-256-GCM)**, so restores require the application keys.

5. DR topology



Element	Primary	DR (warm standby)
App	≥ 3 replicas across 2 AZs; PDB minAvailable: 2; terminationGracePeriodSeconds ≥ 30	Deployment at 0-2 replicas, same image; scaled up on declaration

Element	Primary	DR (warm standby)
Database	HA primary + synchronous standby (automatic AZ failover)	Async read replica (lag < 30 s, alert ALR-91), promotable
CronJobs	Active	Suspended. Never run two journey schedulers, or customers get duplicate contact
Outbound egress	Allow-listed to VETC, TASCO core, Zalo, SMS, voice	Pre-registered DR egress IPs with every partner (part of the IIAs)
DNS	Public hostnames (console, app API, partner API, verify)	GSLB failover record, TTL 60 s

Partner allow-lists are the usual DR blocker. VETC wallet, TASCO core and the SMS provider must allow-list the DR egress IPs **before go-live** (readiness PRC-DR-04).

6. Failover and failback procedures

FO-0 — AZ failure (automatic)

1. The managed Postgres fails over to the synchronous standby (60–120 s). The app's `/health/ready` returns 503 during the switch (RB-01) and the pod pool reconnects.
2. Kubernetes reschedules pods to the healthy AZ; the PDB and HPA keep capacity.
3. L2 verifies: readiness, `GET /api/ops/status`, reconciliation for the window. No customer-facing declaration unless degradation lasts > 15 min.

FO-1 — Region loss / declared disaster (manual, RTO 4 h)

Step	T+	Action	Owner
1	0	Incident Commander declares DR after confirming the primary region is unrecoverable within RTO (provider status, ≥ 30 min outage, or data-centre loss). Decision by the TASCO IT Head (or delegate)	IC
2	0:05	Communications: war room, TASCO/VETC leadership, VETC CS; customer banner (§9)	Comms
3	0:10	Freeze writes at the primary if it is partially alive: scale the primary Deployment to 0, suspend its CronJobs (split-brain prevention)	L2
4	0:15	Promote the DR replica (provider "promote"). Record the last applied LSN/timestamp: this is the actual RPO	DBA
5	0:30	Confirm the DR KMS has every <code>DATA_KEYS</code> key ID, <code>BLIND_INDEX_KEY</code> and the <code>JWT_SECRET</code> , and update the DR secrets: <code>DATABASE_URL</code> → promoted instance	SRE
6	0:45	Scale the DR Deployment to the production replica count; set <code>MIGRATE_ON_START=false</code> (the schema is already current)	SRE
7	1:00	Smoke tests: <code>/health/ready</code> (store:"postgres", db:true), <code>GET /api/audit/verify</code> ok, staff login + MFA, customer session → home → quote, certificate verify, <code>GET /api/ops/status</code> (breakers closed: partner allow-lists working)	L2 + QA
8	1:15	Switch DNS/GSLB to DR (TTL 60 s)	SRE
9	1:30	Run <code>POST /api/ops/jobs/relay</code> (drain pending); events left in processing by the crash are re-claimed after 5 min, otherwise follow RB-06	L2

Step	T+	Action	Owner
10	1:45	Resume CronJobs in DR (marketing touchpoints outside 08:00–20:00 ICT are deferred automatically); reconciliation for the loss window: orders between the last replicated LSN and the disaster may have been paid in VETC without a record here. Reconcile with the VETC settlement file and work any <code>compensation_failed</code> or <code>payment_failed</code> order (RB-11)	L2 + Finance
11	2:00 → 4:00	Monitor SLOs; partner notifications; formal DR status report	IC

FB-1 — Failback to the primary region (planned, low-traffic window)

1. Rebuild the primary-region database as a **replica of the DR primary** (fresh base backup + streaming). Never reuse the old primary's data files (divergence).
2. When the lag is < 5 s, CAB-approved window (after 20:00 ICT): suspend DR CronJobs → scale DR app to 0 (brief outage, ~10 min) → promote the primary-region replica → point secrets → scale up → smoke tests (step 7 above) → DNS back → resume CronJobs.
3. Re-establish the DR replica from the new primary. Reconcile; verify the audit chain.

FO-2 — Logical corruption / bad data change (PITR)

When data is corrupted by a bad job, a bad manual SQL statement or a defect, but the infrastructure is fine:

1. Stop the source of damage (suspend CronJobs, scale writers down if necessary).
2. Restore by **PITR to a new instance** at T– (just before the event).
3. Choose between **(a) full cut-over** to the restored instance (loses all writes after T; only if the window is small) and **(b) surgical repair**: copy the affected rows from the restored instance into production under a change ticket.
4. The audit chain must be verified after any repair. Rows inserted into `audit_log` are never edited; repairs are recorded as new audit entries or in the change ticket.

7. DR test plan and schedule

ID	Test	Scope	Frequency	Success criteria
DR-T1	Backup restore	PITR restore to a new instance in PREPROD at a random point in the last 7 days; app connects; GET <code>/api/audit/verify</code> ok; row counts vs source	Weekly automated (light), quarterly full	Restore ≤ 60 min (Tier 1 budget); data consistent to the target time (RPO ≤ 5 min)
DR-T2	AZ failover	Force a Postgres failover + drain one AZ's nodes under PERF-S1 load (TC-141, TC-142)	Before G3, then semi-annually	Readiness recovers ≤ 2 min; no lost committed orders; 5xx limited to the switch window
DR-T3	Region failover drill	Full FO-1 in PREPROD/DR (or PROD with a planned window after W1)	Before W1 (March 2027), then annually	RTO ≤ 4 h, RPO ≤ 5 min demonstrated; partner calls succeed from DR egress IPs
DR-T4	Rebuild from code	New cluster from IaC + image registry + secrets escrow	Annually	Platform up from scratch ≤ 8 h
DR-T5	Key recovery	Recover <code>DATA_KEYS</code> (all IDs) and <code>BLIND_INDEX_KEY</code> from escrow; decrypt a sample in an isolated environment	Semi-annually	Two-person procedure completes ≤ 1 h
DR-T6	Vendor outage tabletop	Walk through §8 scenarios with VETC CS, the telesales supervisor and the voice vendor	Before pilot, then semi-annually	Roles clear; templates approved; contact lists valid

ID	Test	Scope	Frequency	Success criteria
DR-T7	Outbox recovery	Kill pods mid-relay; recover stuck processing and dead letters per RB-06	Before G3	No lost or duplicated customer messages beyond the documented caveats

Window	Tests
January 2027 (UAT week T3)	DR-T1, DR-T2, DR-T5, DR-T6, DR-T7 (gate G3 evidence)
March 2027 (before W1)	DR-T3 full region failover
Every quarter	DR-T1 full
Semi-annual	DR-T2, DR-T5, DR-T6
Annual	DR-T3, DR-T4

Every DR test produces a report: date, scenario, actual RTO/RPO, issues, actions, sign-off by the TASCO IT Head.

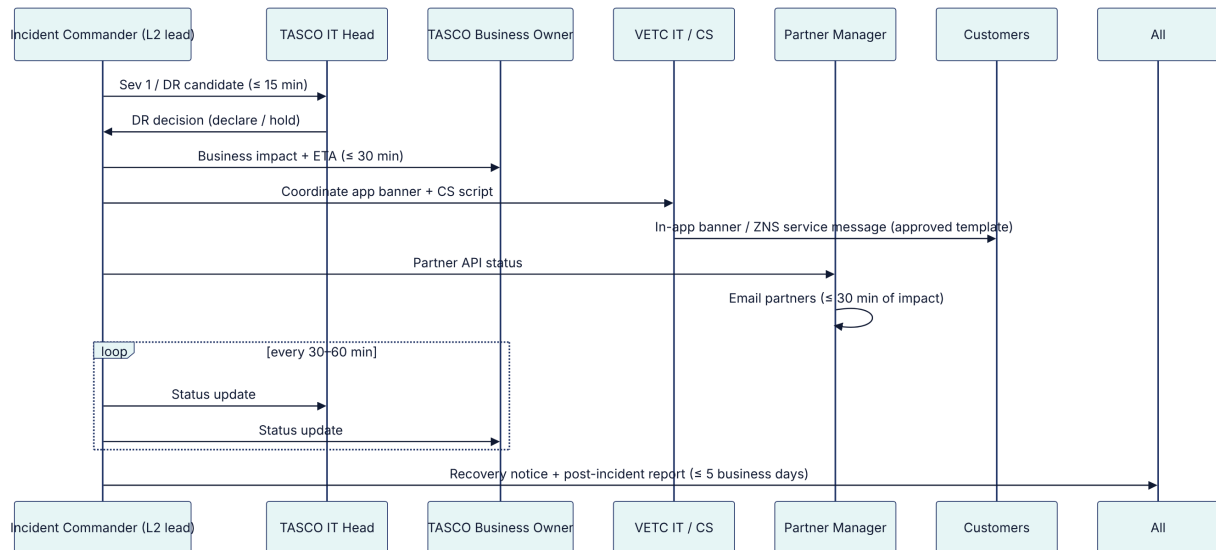
8. Business continuity: telesales and voice vendor outage

Scenario	Detection	Continuity actions	Customer impact
Voice-AI vendor outage (circuit voice-ai, RB-05)	ALR-10, voice_calls_total flat	1. Stop launching /api/voice/campaign. 2. Journeys: voice_bot steps fail. The fallback is a rule change (maker-checker) adding telesales to the voice step's channels, or raising hot-lead prioritisation for agents (GET /api/leads? tier=hot). 3. Supervisors add agent shifts. 4. Digital steps (push/ZNS/SMS) continue	Fewer proactive calls; digital reminders continue
Telesales unavailable (call centre outage, strike, telephony down)	Supervisor report; ALR-83 unworked handoffs	1. Pause escalation to telesales: rule change removing telesales from journey steps, or temporarily suspend journeys (SOP-07) if there is no other channel. 2. The voice bot outcome hot_handoff still creates handoffs; switch the bot outcome guidance to link_sent (rule change in content.voicebot; customers receive the in-app link). 3. Partner channel unaffected	Hot leads served by self-service link
Zalo ZNS or SMS provider outage	ALR-31, RB-04	Channel fallback is built in (app push → ZNS → SMS order per step). For a prolonged outage, a rule change reorders channels. Customers with no app and no other channel wait for recovery	Some reminders delayed
VETC wallet outage	RB-02	No alternative payment method in scope (by design: customers pay only in the official app, staff never take payment). Banner in the app; telesales collect intent, send quotes and call back; journeys paused (SOP-07) so reminders do not drive customers into failures	Purchases delayed
TASCO core outage	RB-03	Stop sales (pause journeys, hide the renew banner); queue handoffs for callback; orders that fail at issuance are compensated automatically (lines cancelled, full refund); any compensation_failed order is worked manually (RB-11)	Purchases delayed
VETC SSO / app outage	VETC notification	Zalo mini-app path (signed links) continues if Zalo is up; telesales assist	Partial

Scenario	Detection	Continuity actions	Customer impact
Platform total outage (DR invoked)	§6	VETC CS scripts; TASCO hotline for claims FNOL; partners informed	Per RTO

Manual workaround register (kept by L2): approved customer-facing templates per scenario (vi/en), CS scripts, partner notice template, decision log template.

9. Communication plan



Audience	Message owner	Channel	Timing
Executive (TASCO IT Head, Business Owner, VETC IT Lead)	Incident Commander	Phone + email	Within 30 min of declaring Sev 1 / DR; then hourly
Staff users (telesales, campaign, claims)	Service desk	Teams/Zalo broadcast	Within 30 min
Customers	VETC CS + TASCO Marketing (Compliance-approved templates)	In-app banner, ZNS service message, hotline IVR	When the purchase path is down > 30 min
Partners	Partner Manager	Partner status email / portal	Within 30 min of partner API impact
Regulators / data subjects (personal data breach)	DPO with Legal	Per statutory process	Per legal deadlines

10. Plan maintenance

- Reviewed after every DR test, every Sev 1, and every architecture change (new region, provider, integration).
- Contact lists verified quarterly.
- Owners: SRE (technical procedures), TASCO IT Head (BIA and tiers), VETC CS (customer communications), Partner Manager (partner communications).